

Compustat Completeness and Expanded Decisions

Goal

The simulation should generate a quarterly panel that covers the most important Compustat Quarterly variables. This means the simulated world must create enough economic activity to populate these columns meaningfully -- not as afterthoughts, but as natural consequences of firm decisions and market outcomes.

Gap Analysis: Current Panel vs. Real Compustat

Variables Already Covered (40 columns)

Our current panel covers the core of Compustat Quarterly well:

Income Statement: saleq, cogsq, gpq, xrdq, xsgaq, dpq, oiadpq, xintq, piq, txtq, niq

Balance Sheet -- Assets: cheq, rectq, invtq, ppentq, atq

Balance Sheet -- Liabilities: apq, acoq, txpq, dlcq, lctq, dlittq, ltq

Balance Sheet -- Equity: cstq, ceqq, req

Cash Flow: oancfq, ivncfq, fincfq, chechq, capxq

Equity & Market: sstkq, prstkq, dvq, prccq, cshoq, mkvaltq

Custom: default_flag, pricing_error

Variables Missing (Important Compustat Fields)

Variable	Compustat Name	What It Requires	Priority
Goodwill	gdwlq	M&A transactions (now sup	High
Goodwill impairment	gdwlipq	Impairment testing post-a	High
Acquisition amount	aqaq	M&A transactions	High
Intangible assets	intanq	R&D capitalization regime	High
Other income/expense	nopiq / spiq	Gains/losses on asset sal	Medium
Accounts receivable (gros	rectrq	Credit losses / bad debt	Medium
Allowance for doubtful ac	--	Bad debt experience	Medium
Prepaid expenses	xppq	Advance payments for raw	Low
Other current assets	acoq (asset side)	Miscellaneous current ass	Low
Deferred revenue	drcq	Advance patient payments	Medium
Other long-term liabiliti	loq	Pension, lease obligation	Medium
Additional paid-in capita	apicq	Already tracked internall	High
Treasury stock	tstkq	Buyback accounting	High
Accumulated OCI	aociq	Fair value regime, pensio	Medium
Revenue per segment	--	Geographic or product-lin	Low
Employees	empq	Workforce tracking (alrea	Medium
Capital lease obligations	--	If facility leasing is mo	Low
Pension/retirement expens	--	If workforce benefits are	Low
Earnings per share	epsfxq / epsfiq	Computed from NI and shar	High
Book value per share	bkvlpq	Computed from equity and	High
Dividends per share	dvpsxq	Computed from dividends a	High

Operating lease expense	--	Facility costs	Low
Restructuring charges	rcq	Post-acquisition integrat	Medium

Expanded Decision Set

To populate the missing variables, firms need to make richer decisions and the world needs to generate richer events:

New Firm Decisions

Decision	Populates	Description
Workforce plan	empq, restructuring	Hiring target, layoff decision, co
Working capital policy	rectq, invtq, apq, drcq	Collection aggressiveness, invento
Debt management	dlcq, dlittq	Repayment schedule, refinancing re
M&A proposal	gdwlq, aqaq, intanq	Acquisition offer (see doc 10)
Asset disposal	nopiq, spiq	Sell idle equipment, close a facil
Lease vs. buy	loq	Lease facilities instead of buildi
Reserves and provisions	acoq (liability), loq	Litigation reserves, warranty prov

Expanded Firm Decision JSON

```
{
  "price": 92000,
  "production": 220,
  "capex": 20000000,
  "rd_spend": 30000000,
  "rd_allocation": {"product": 0.55, "process": 0.25, "delivery": 0.20},
  "sga_spend": 15000000,

  "workforce": {
    "hiring_target": 25,
    "layoff_count": 0,
    "avg_compensation_increase_pct": 3.0
  },

  "working_capital": {
    "collection_aggressiveness": "normal",
    "inventory_target_quarters": 1.5,
    "supplier_payment_terms": "net_45"
  },

  "debt_management": {
    "term_debt_repayment": 0,
    "refinancing_request": false
  },

  "asset_disposal": {
    "dispose_assets": false,
    "assets_to_dispose": []
  },

  "provisions": {
```

```
    "litigation_reserve_addition": 2000000,
    "warranty_reserve_addition": 500000
  },

  "ma_proposal": null,

  "equity_issuance_request": 0,
  "debt_request": 0,
  "dividends": 0,
  "buybacks": 0,

  "reasoning": "...
}
```

Expanded Environment Outcomes

The environment now also generates:

```
{
  "firm_outcomes": [
    {
      "firm_id": "firm_0",
      "units_sold": 19200,
      "market_share": 0.217,
      "bad_debt_pct": 0.02,
      "patient_lawsuits_files": 1,
      "employee_departures": 3,
      "supplier_issues": "none",
      "special_items": []
    }
  ],
  "industry_events": [
    {
      "type": "restructuring",
      "affected_firm": "firm_3",
      "description": "Firm 3 announced facility closure and 15% workforce reduction",
      "restructuring_charge": 25000000
    }
  ]
}
```

Expanded Compustat Panel (Final Column List)

Income Statement (18 columns)

Column	Name	Source
saleq	Net sales	units_sold * price
cogsq	Cost of goods sold	units_sold * unit_cost
gpq	Gross profit	saleq - cogsq
xrdq	R&D expense	actual_rd_spend (or net of capital
xsgaq	SGA expense	actual_sga + compensation costs

dpq	Depreciation & amortization	PPE depreciation + intangible amor
gdwlpq	Goodwill impairment	From impairment test (if M&A)
rcq	Restructuring charges	Layoffs, facility closure
oiadpq	Operating income	gpq - xrdq - xsgaq - dpq - gdwlpq
nopiq	Non-operating income	Asset sale gains/losses, one-time
xintq	Interest expense	On all debt
spiq	Special items	Unusual one-time events
piq	Pretax income	oiadpq + nopiq - xintq + spiq
txtq	Tax expense	max(0, piq * tax_rate) with loss c
niq	Net income	piq - txtq
epsfxq	EPS (fully diluted)	niq / diluted_shares
epsfiq	EPS (basic)	niq / shares_outstanding
dvpsxq	Dividends per share	dvq / cshoq

Balance Sheet -- Assets (14 columns)

Column	Name	Source
cheq	Cash	Settlement residual
rectq	Accounts receivable (net)	theta_AR * saleq - allowance
invttq	Inventories	Unsold units * unit_cost
xppq	Prepaid expenses	Raw material deposits
acoq_a	Other current assets	Miscellaneous
actq	Total current assets	Sum of above
ppentq	PP&E (net)	Gross PPE - accum depreciation
intanq	Intangible assets	Capitalized R&D + acquired IP
gdwlq	Goodwill	From M&A (purchase price - fair va
aoq	Other long-term assets	Miscellaneous
atq	Total assets	actq + ppentq + intanq + gdwlq + a

Balance Sheet -- Liabilities (13 columns)

Column	Name	Source
apq	Accounts payable	theta_AP * cogsq
acoq	Accrued expenses	Wages, utilities, other accruals
txpq	Taxes payable	Current quarter tax due
drcq	Deferred revenue	Advance patient payments (if any)
dlcq	Current debt	Revolver balance + current portion
lctq	Total current liabilities	Sum of above
dlttq	Long-term debt	Term debt (non-current portion)
loq	Other long-term liabilities	Litigation reserves, provisions, l
ltq	Total liabilities	lctq + dlttq + loq

Balance Sheet -- Equity (8 columns)

Column	Name	Source
cstkq	Common stock (par)	Shares * par value
apicq	Additional paid-in capital	IPO + secondary proceeds above par
req	Retained earnings	Cumulative NI - cumulative dividen
tstkq	Treasury stock	Cumulative buybacks at cost
aociq	Accumulated OCI	Fair value adjustments (if FV regi
ceqq	Total common equity	cstkq + apicq + req - tstq + aoci

Cash Flow Statement (8 columns)

Column	Name	Source
oancfq	Operating cash flow	NI + depreciation + WC changes + n
ivncfq	Investing cash flow	-capex - acquisitions + asset sale
fincfq	Financing cash flow	Equity + debt - repayments - div -
chechq	Change in cash	oancfq + ivncfq + fincfq
capxq	Capital expenditure	Actual capex posted
aqaq	Acquisition spending	Cash paid for acquisitions
sstkq	Stock issuance	Equity raised this quarter
prstkq	Stock repurchase	Buybacks this quarter
dvq	Dividends paid	Cash dividends this quarter

Market & Other (10 columns)

Column	Name	Source
prccq	Stock price	Set by investment bank
cshoq	Shares outstanding	
mkvaltq	Market capitalization	prccq * cshoq
bkvlpq	Book value per share	ceqq / cshoq
empq	Employees	From workforce tracking
default_flag	Default indicator	
acquiree_flag	Was acquired this quarter	
acquirer_flag	Made acquisition this quarter	
pricing_error	P - P*	For research
run_id / firm_id / incarnation / f	Keys	

Total: ~76 columns (vs. 44 previously)

Validation Invariants (Updated)**Hard Invariants**

All previous invariants remain. New additions:

- $atq == actq + ppentq + intanq + gdwq + aoq$ (asset decomposition)
- $lctq == apq + acoq + txpq + drcq + dlcq$ (current liabilities decomposition)
- $ltq == lctq + dlittq + loq$ (total liabilities decomposition)
- $ceqq == cstq + apicq + req - tstq + aociq$ (equity decomposition)
- $gdwq \geq 0$ (goodwill cannot be negative)
- $gdwlipq \geq 0$ (impairment is a loss, recorded as positive expense)
- $empq \geq 0$ (non-negative employees)
- $epsfxq == niq / diluted_shares$ (within rounding)

Soft Invariants

- $empq$ should track roughly with revenue and SGA trends
- $gdwq$ only changes on acquisitions or impairments
- $intanq$ only changes under R&D capitalization regime
- $aqaq > 0$ only in quarters with actual acquisitions

What Drives Each Compustat Variable

This mapping ensures that every column has an explicit economic mechanism:

Variable	Driven By
saleq	Demand system (environment) * firm's price
cogsq	Production volume * unit cost (driven by process R&D)
xrdq	Firm's R&D decision (clamped by feasibility)
xsgaq	Firm's SGA decision + workforce compensation
dpq	PPE depreciation schedule + intangible amortization
gdwlipq	Annual goodwill impairment test
rcq	Firm's restructuring decision + environment events
xintq	Outstanding debt * interest rates (set by banks)
txtq	Tax code applied to pretax income
cheq	Residual from all cash flows
rectq	Revenue * collection cycle - bad debt
invttq	Production - sales, valued at unit cost
ppentq	Prior PPE + capex - depreciation
intanq	Capitalized R&D + acquired intangibles - amortization
gdwlq	Acquisition premium - impairments
apq	COGS * payment cycle
acoq	Wages payable + accrued expenses
dlcq	Revolver balance + LTD current portion
dlttq	Term debt issued - repayments - reclassification to
loq	Litigation reserves + lease obligations + provision
prccq	Investment bank valuation
empq	Firm workforce decisions + environment attrition